Ishan Gutpa

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Mechanical Engineering — Dec 2026
Courses: Modern Control Theory, ML/AI for Engineers, Intro to Robot Learning — GPA: 3.76/4.00

The University of Texas at Austin — Austin, TX
Bachelors of Science in Mechanical Engineering | Robotics Minor | Certificate in Computer Science — May 2025
Courses: Systems and Controls, Mechatronics, ML for Engineers, Intro to Robotics — GPA: 3.88/4.00

SKILLS

Programming: Python, Matlab, Simulink, PyTorch, Git, C++, Scipy, Sklearn, PyBullet, Gymnasium
Software Tools: ROS2, CARLA, Mujoco, Webots, Solidworks, Arduino, Excel, Powerpoint, Blender

ACADEMIC PROJECTS

F1Tenth Autonomous Racing / Pittsburgh, PA — Jan 2026 - Present

- Developed ROS2-based perception and autonomy modules for an F1TENTH vehicle, integrating LiDAR processing, localization, and closed-loop control for indoor autonomous racing.
- Implemented LiDAR-based SLAM/localization to support GPS-denied navigation, enabling map-based autonomous driving and real-time pose estimation on indoor tracks.
- Built and tuned LiDAR-driven obstacle avoidance behaviors including wall-following and follow-the-gap, improving navigation robustness under constrained geometry and noisy range measurements.

Learning-Based Behavior Planning for Urban Driving / Pittsburgh, PA — Jan 2026 - Present

- Developing an RL-based behavior planner in CARLA for unprotected left turns and merges, benchmarked against the simulator's rule-based BehaviorAgent baseline.
- Designing observation and reward formulations to reason over ego state, nearby traffic, safety margins, and uncertain interaction outcomes, while interfacing the learned planner with low-level trajectory following/control.
- Evaluating performance using success rate, collision rate, completion time, and behavior under varying traffic density/aggressiveness, with emphasis on robustness to uncertain and dynamic driving environments.

Simulated Autonomous Vehicle Control / Pittsburgh, PA — Oct 2025 - Dec 2025

- Developed PID and pole placement algorithm to autonomously control a car around a track with an average of 0.47 meters discrepancy from the trajectory in Webots.
- Improved the controls using LQR algorithm and A star for remapping around a moving car, reducing time of completion by 20%.
- Implemented EKF-based state estimation / SLAM for autonomous navigation, enabling localization and trajectory tracking with average path deviation of 0.7 m.
- Designed an MRAC with a baseline LQR controller to fly a quadrotor with up to 68% loss of thrust in one motor.

PROFESSIONAL EXPERIENCE

Pike Robotics / Robotics Engineer Intern / Austin, TX — Jan 2024 - Aug 2024

- Won $120K grant; led the creation of a project proposal presentation including BOM, initial CAD conceptual design, and short animation using Solidworks and Blender to convey our suggested solution better.
- Created a mechanical multiplexer ejection system, resulting in a scalable and cheap solution that minimizes motors needed.
- Optimized sensor mount through repeated design iterations, resulting in increased sensor coverage and reliability.

Pike Robotics / Product Engineer Intern / Austin, TX — Feb 2023 - Dec 2023

- Created an actuated wedge system and tested lifting force data, decreasing inspection setup time by 10%.
- Pioneered a novel two-robot system direction for the company by conceptualizing anchor systems & a companion robot to manage a tether, leveraging Solidworks & additive manufacturing.
- Improved DFM/A on anchor subassembly, through clip on design, hence reducing assembly time by 30%.